

mRSubLoc: A novel multi-label learning framework integrating RNA large language model for mRNA subcellular localization

Xiao Wang, Lixiang Yang, Rong Wang, Yongfeng Zhang

Abstract—The subcellular localization of messenger RNA (mRNA) is essential for the regulation of gene expression and plays a pivotal role in targeted drug development. Although several computational models have been developed to predict mRNA localization, these approaches still face challenges in sequence representation and exhibit limited performance in handling multi-localization tasks. In this paper, we propose mRSubLoc, a novel multi-label deep learning framework for predicting mRNA subcellular localization. The model integrates the RNA large language model RNAErnie with one-hot encoding and Word2Vec embeddings to construct a comprehensive representation of mRNA sequences. A text convolutional neural network (TextCNN) is employed to capture local feature patterns, while a bidirectional long short-term memory network (BiLSTM) is used to capture long-range dependencies. These features are fused using a multi-head self-attention mechanism to effectively capture localization-specific characteristics. Finally, a multi-layer perceptron (MLP) explores complex dependencies among multiple localization sites, facilitating accurate mRNA subcellular localization prediction. Experimental results on a testing set demonstrate that mRSubLoc significantly outperforms state-of-the-art methods across multiple metrics, including Aiming (0.7858), Coverage (0.6212), Accuracy (0.6161), Absolute-True (0.3070), and Absolute-False (0.1319). This study proposes a novel approach for predicting mRNA subcellular localization and provides new perspectives for advancing disease diagnosis and drug discovery in biomedical research.

Index Terms—Deep learning; Large language model; subcellular localization of mRNA; Multi-label classification; Self-attention mechanism.

I. INTRODUCTION

MESSENGER RNA (mRNA) is synthesized through the transcription process, using a single strand of DNA as a template [1]. The distribution of mRNA across cellular compartments is closely related to numerous physiological processes, such as the cell polarity, embryonic development, synaptic plasticity, and stress

responses [2]. The 3' untranslated region (UTR) and 5' cap structure of mRNA typically contain essential signals that mediate mRNA localization. These signals allow mRNA to be precisely targeted to cellular structures such as the membrane, cytoplasm, and nucleolus, ensuring that protein synthesis occurs in the correct subcellular compartment [3]. Impairment of this localization mechanism has been associated with the onset of diverse diseases, such as cancer, neurodegenerative diseases, and developmental defects [4]. Therefore, gaining a holistic understanding of the mRNA localization mechanism is biologically significant and crucial for targeted drug therapy.

Currently, traditional experimental methods, including fluorescence in situ hybridization (FISH) [5], APEX-RIP [6], and CeFra-seq [7], provide valuable insights into RNA localization within cells. However, these techniques are not only consuming time and intensive resource but also struggle to handle large scale data [8]. To address these limitations, computational approaches based on machine learning and deep learning have gradually become indispensable complementary tools, which efficiently process massive RNA localization datasets and significantly enhance analytical precision and efficiency. To address mRNA subcellular localization prediction, Yan et al. initially introduced RNATracker [9], which employs recurrent neural networks (RNNs) to integrate mRNA sequence and secondary structure features. Subsequently, Zhang et al. proposed iLoc-mRNA [10], which extracts k-mer features, applies ANOVA with a binomial distribution for feature selection, and uses a support vector machine (SVM) to predict mRNA subcellular localization. Garg et al. proposed mRNALoc [11] based on a machine learning approach. Li et al. introduced SubLocEP [12], which forms a dual-layer architecture for mRNA subcellular localization by applying a 3:2 weighted combination of sequence-based physicochemical properties. Subsequently, Yuan et al. proposed RNALight [13], a LightGBM-based model aimed at identifying nucleotide k-mers associated with mRNA subcellular localization. These methods are limited to identifying a single subcellular location for mRNA.

This work was supported by the Key Research Project of Colleges and Universities of Henan Province (No.22A520013, No.23B520004), the Key Science and Technology Development Program of Henan Province (No.232102210020, No.252102210137), and the Training Program of Young Backbone Teachers in Colleges and Universities of Henan Province (No.2019GGJS132) (Corresponding authors: Xiao Wang).

Xiao Wang is with the Henan Provincial Key Laboratory of Data Intelligence for Food Safety and the School of Computer Science and Technology, Zhengzhou University of Light Industry, Zhengzhou 450000, China (e-mail: wangxiao@zzuli.edu.cn).

Lixiang Yang is with the School of Computer Science and Technology, Zhengzhou University of Light Industry, Zhengzhou 450000, China (e-mail: yanglixiang@zzuli.edu.cn).

Rong Wang is with the School of Electronic Information, Zhengzhou University of Light Industry, Zhengzhou 450000, China (e-mail: wangrong@zzuli.edu.cn).

Yongfeng Zhang is with the School of Computer Science and Technology, Zhengzhou University of Light Industry, Zhengzhou 450000, China (e-mail: zhangyongfeng@zzuli.edu.cn).

Availability and implementation: Source code and datasets are publicly available at <https://github.com/YangLX-Coder/mRSubLoc>.

However, with an enhanced understanding of the complexity of mRNA subcellular localization, it is now clear that mRNA is not restricted to a single compartment but can simultaneously localize to multiple subcellular localizations. Wang et al. proposed DM3Loc [14], a method that first encodes the sequence using one-hot encoding and then utilizes a multi-head self-attention mechanism for prediction. Subsequently, Bi et al. developed Clarion [15], a model that uses a weighted sequence method for multi-label transformation to predict multiple subcellular locations of mRNA. Choudhury et al. developed MRSLpred [16], a hybrid model which combines XGBoost-based sequence composition analysis with motif-based alignment techniques to achieve fast and accurate mRNA localization prediction at the genome scale. Subsequently, Zuo et al. introduced MSlocPRED [17], which leverages deep transfer learning for multi-label prediction of mRNA subcellular position, integrating the MDNDO-SMDU resampling strategy. Zhang et al. developed MBSCLoc [18], a model which applies multi-class contrastive learning and cluster-balanced subspace partitioning, and integrates multiple XGBoost classifiers to address data imbalance and improve localization prediction. These computational approaches have served as effective tools and valuable resources for advancing research on mRNA subcellular localization. Nevertheless, these methods usually only consider the physical and chemical properties of the mRNA sequence to represent it, or only use a single language model to embed the sequence. These methods may not fully reflect the complex and ambiguous patterns and relationships within the sequence. Additionally, extracting localization-specific features for each subcellular compartment and learning the relationships between location labels are crucial for multi-label subcellular localization tasks.

In this paper, we develop mRSubLoc, an innovative approach for multi-label prediction of mRNA subcellular localization. To enhance the representation of biological sequences, we introduced the RNA large language model RNAErnie [19] to further strengthen the feature expression of mRNA sequences. We combined one-hot encoding, Word2Vec embeddings [20], and RNAErnie features for the multi-localization prediction of mRNA across subcellular compartments. We adopted TextCNN [21] and BiLSTM [22] to separately learn the feature representations for each type of input. The features extracted by TextCNN and BiLSTM are then passed through a multi-head self-attention mechanism, which identifies the optimal features for each subcellular location, enabling effective representation of the overall subcellular location pattern of the mRNA sequences. The subcellular location distributions obtained from one-hot encoding, RNAErnie features, and Word2Vec embeddings are then concatenated to form a new unified distribution. This distribution is subsequently refined and further analyzed by a multi-layer perceptron (MLP), which captures the complex interdependencies among the locations. The results of 5-fold cross-validation on the training set indicate that, compared to traditional methods, RNA large language models demonstrate a more effective capability in representing mRNA sequences. Additionally, the use of a multi-head self-attention mechanism to select the optimal features at each position, which refer to the most informative and localization-specific representations that

contribute most significantly to distinguishing between different subcellular locations, and the adoption of a multi-label learning strategy to learn the relationships between subcellular locations, significantly enhances the model's capability. Furthermore, the results from the testing set indicate that mRSubLoc surpasses other state-of-the-art approaches across most performance metrics.

II. METHODS

A. Dataset

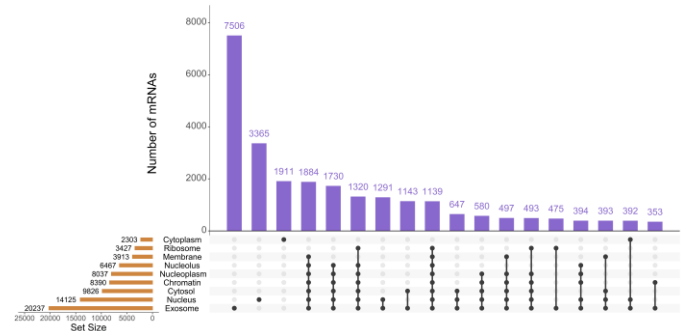


Fig. 1. Upset plot showing the intersections of mRNAs across nine subcellular localizations in the benchmark dataset.

Constructing a precise and valid dataset is a critical step in the development of an effective predictive model. In this paper, we employed a preprocessed dataset provided by MSlocPRED [17], which includes data from nine distinct subcellular compartments. All annotations of subcellular location and corresponding mRNA sequences in this dataset are obtained from the RNALocate v2.0 [23] database. To reduce the influence of redundant sequences from the RNALocate v2.0 database on the model's performance, CD-HIT-EST [24] was applied to eliminate sequences with over 80% homology, maintaining a similarity threshold of below 80% between any two nucleotide sequences. When an mRNA possesses multiple localization annotations, they are combined and assigned as multiple labels. To ensure data balance and statistical reliability in multi-label learning, label combinations with fewer than 300 sequences were removed, as these rare label combinations may lead to overfitting and hinder the model's ability to generalize. In addition, duplicate sequences were eliminated. A total of 25,513 mRNAs were ultimately acquired and used to establish the benchmark dataset. Fig. 1 presents the intersections of mRNAs across nine subcellular localizations, along with the number associated with each localization as follows: exosome (20,237), nucleus (14,125), nucleoplasm (8037), chromatin (8390), cytoplasm (2303), nucleolus (6467), cytosol (9826), membrane (3913), and ribosome (3427).

Subsequently, guided by the statistical information shown in Fig. 1, the benchmark dataset was split into training and testing sets at a 9:1 ratio by stratified sampling for model construction and evaluation. The distribution of subcellular localization types in both the training and testing sets is summarized in Table 1. The training set was employed to construct and optimize the subcellular localization prediction model. The testing set was reserved for independent performance evaluation, allowing a fair comparison with existing state-of-the-art models.

TABLE I

STATISTICS OF THE TRAINING AND TESTING SETS

Subcellular	Training set	Testing set
Exosome	18,214	2023
Nucleus	12,720	1405
Nucleoplasm	7248	789
Chromatin	7567	823
Cytoplasm	2060	243
Nucleolus	5828	639
Cytosol	8841	985
Membrane	3519	394
Ribosome	3091	336

B. mRSubLoc architecture

The overall architecture of mRSubLoc, as illustrated in Fig. 2, comprises six components: (A) dataset construction; (B) sequence embedding; (C) localization-specific feature extraction via multi-head self-attention; (D) relationship learning between localization labels; (E) prediction module; and (F) performance evaluation.

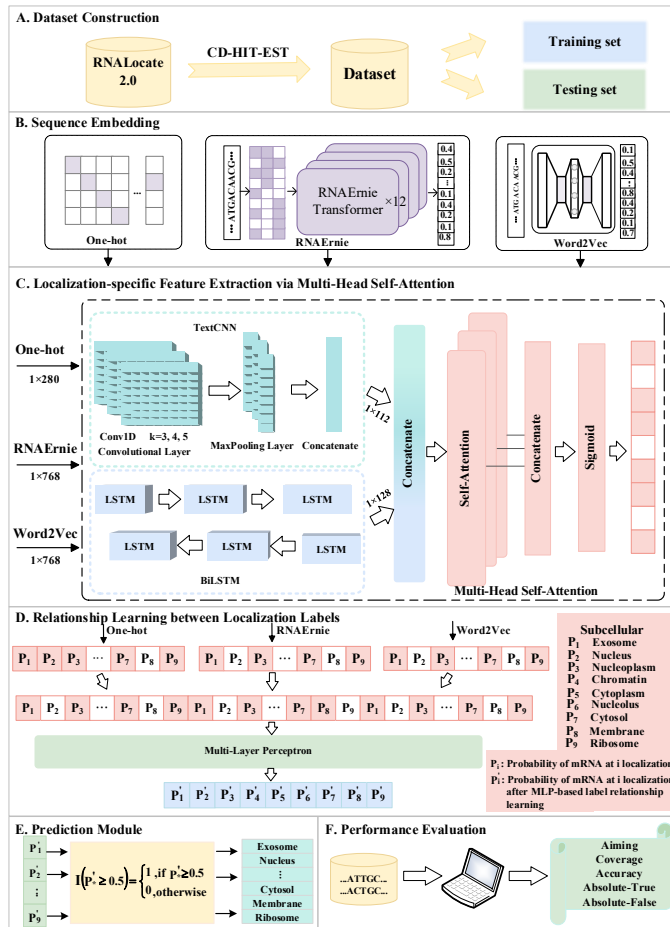


Fig. 2. The framework of mRSubLoc.

C. Subcellular localization-specific feature extraction

In multi-label learning tasks, effectively extracting class-specific features is essential for improving model performance, as different classes exhibit distinct feature distributions [25]. In the context of subcellular location prediction, each location corresponds to a unique molecular environment and feature

pattern, necessitating a mechanism capable of capturing location-specific information. To address this, we employ a multi-head self-attention mechanism to compute global attention weights, enhancing key features relevant to specific locations while suppressing irrelevant information. Subsequently, based on the prior feature distributions of subcellular locations, the attention outputs are selectively filtered and aggregated to form distinct representations for each location. This approach improves the model's ability to capture localization-specific differences, enhances classification accuracy, and provides more discriminative feature representations for downstream bioinformatics analyses. The pseudo-code for the subcellular localization-specific feature extraction process is presented in Algorithm 1.

Algorithm 1 The Process of Extracting Subcellular Localization-Specific Features

Input: Projection matrices $W_Q, W_K, W_V \in \mathbb{R}^{D \times d_k}$, output matrix W_O , subcellular locations $S = 9$, feature aggregation function $f(\cdot)$.

Output: Extracted features $\{F_{S_1}, \dots, F_{S_9}\}$.

1. Initialize projection matrices W_Q, W_K, W_V, W_O .
2. Set number of subcellular locations: $S = 9$.
3. Define feature aggregation function $f(\cdot)$ as applying dropout with rate 0.3 followed by a fully connected layer: $f(\cdot) = fc(dropout_{0.3}(\cdot))$.
4. **for** each input $X = \mathbb{R}^{B \times L \times D}$ **do**
5. Compute Query, Key, and Value matrices:
6. $Q = XW_Q, K = XW_K, V = XW_V$.
7. Compute attention scores: $A = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)$.
8. Apply attention: $Z = AV$.
9. Output projection: $X_{out} = ZW_O$.
10. **For** each location $S_i, i = 1$ to 9 **do**
11. Extract relevant features X_{S_i} from X_{out} .
12. Compute location-specific feature representation $F_{S_i} = f(X_{S_i})$.
13. **return** $\{F_{S_1}, \dots, F_{S_9}\}$.

D. Learning the interdependence among localization labels

The coordinated activity of mRNA across distinct subcellular compartments is often associated with intricate biological regulatory mechanisms [2]. These biological dependencies suggest that subcellular localization outcomes tend to co-occur and exhibit interdependent patterns. To improve prediction accuracy, it is essential to learn the interdependencies between subcellular localization labels to uncover their potential co-occurrence patterns. As shown in Algorithm 2, we employ a multilayer perceptron (MLP) to learn the interdependences between different location labels, thereby optimizing the prediction probabilities for subcellular localization. The MLP utilizes nonlinear transformations and fully connected layers to extract features from the input subcellular localization probabilities, capturing higher-order correlations between different localization labels. Specifically, each updated probability P'_i incorporates information not only from its

corresponding label P_i , but also from other related labels P_j via the learned interdependency weights α_{ij} . This process adaptively recalibrates the feature representation for each label by embedding its co-localization context, thereby enabling the model to extract biologically meaningful, label-specific latent dependencies. Ultimately, through the relationship learning process in the MLP, the accuracy of subcellular localization prediction is enhanced.

Algorithm 2 The Process of Learning Interdependencies among Localization Labels

Input: Localization labels $\{P_1, \dots, P_9\}$ with semantic meanings {Exosome, Nucleus, Cytoplasm, ...}

Output: Refined probabilities $\{P'_1, \dots, P'_9\}$ after relationship learning

1. Encode each label P_i through three parallel pathways:
 $P_i^{\text{One-hot}} \in \mathbb{R}^d$ $P_i^{\text{Word2Vec}} \in \mathbb{R}^d$ $P_i^{\text{RNAErnie}} \in \mathbb{R}^d$
 2. Concatenate embeddings:
 $h_i^{(0)} = [P_i^{\text{One-hot}} || P_i^{\text{Word2Vec}} || P_i^{\text{RNAErnie}}]$
 3. Define MLP with L hidden layers:
 $h_i^{(l)} = \sigma(W^{(l)} h_i^{(l-1)} + b^{(l)}) \quad \forall l \in \{1, \dots, L\}$
 where $\sigma = \text{ReLU}$ activation, $W^{(l)} \in \mathbb{R}^{d_h \times d_h}$
 4. Compute interdependency weights:
 $\alpha_{ij} = \text{softmax}(h_i^{(L)} \cdot h_j^{(L)})$
 5. Refine logits via learned relationships:
 $\hat{Z}_i = \sum_{j=1}^9 \alpha_{ij} h_j^{(L)} \quad \forall i \in \{1, \dots, 9\}$
 6. Optimize parameters using BCEWithLogitsLoss:
 $\mathcal{L} = - \sum_{i=1}^9 w_i [y_i \log \sigma(\hat{Z}_i) + (1 - y_i) \log(1 - \sigma(\hat{Z}_i))]$
 7. **return** $\{P'_1, \dots, P'_9\}$
-

E. Input and embedding of mRNA sequences

1) Large language model

Advancements in the field of natural language processing have encouraged researchers to leverage large language models to simulate and analyze nucleotides in trillions of RNA sequences, such as RNABERT [26], RNA-MSM [27], RNA-FM [28], and RNAErnie. Among these methods, RNABERT emphasizes base-level embeddings aimed at RNA structural alignment and clustering, which may be less focused on functional localization tasks. RNA-MSM models evolutionary information from homologous sequences, but RNA's relatively low sequence conservation can limit its representation of functional diversity related to subcellular localization. RNA-FM, trained on large-scale non-coding RNA data, primarily focuses on capturing general characteristics of ncRNAs. In contrast, RNAErnie integrates sequence context with biological priors such as motifs and RNA types, providing comprehensive and biologically relevant features well suited for mRNA subcellular localization prediction.

In this paper, RNAErnie is utilized for feature representation in the prediction of mRNA subcellular localization. RNAErnie is constructed designed with the transformer architecture, which processes RNA sequence data through multi-head self-attention mechanisms [19]. This enables the model to attend to different regions of the sequence at the same time and adjust the weights dynamically. With a hidden state dimension of 768, the model can capture complex patterns in RNA sequences,

including long-range dependencies and local interactions. This architectural design provides RNAErnie with a robust foundation, enabling it to deeply understand the intrinsic structure and functional characteristics of RNA sequences.

To strengthen the expressive capacity of RNA sequence features, RNAErnie employs multiple masking strategies, including base-level masking, subsequence-level masking, and motif-level random masking. These masking strategies empower the model to concurrently capture both proximal and distal dependencies within RNA sequences, thereby facilitating a more nuanced and in-depth characterization of their inherent complexity. Additionally, RNAErnie incorporates RNA type tokens by appending different types of RNA (such as mRNA, miRNA, etc.) as special vocabularies to the end of sequences. This design allows the model to recognize features specific to different RNA types, thereby enhancing its adaptability and generalization across various downstream tasks.

In the task of predicting mRNA subcellular localization, these design choices of RNAErnie enable it to effectively model the complexity of mRNA sequences. Through the transformer architecture and multi-level masking strategies, the sequence embeddings generated by RNAErnie not only contain rich structural information but also include functional patterns related to subcellular localization. These embeddings function as discriminative feature representations within subsequent classification architectures, thereby significantly enhancing the predictive performance for mRNA subcellular localization.

2) One-hot encoding

In bioinformatics, one-hot encoding is a standard technique for converting genetic sequences into machine-readable representations [29]. It encodes categorical variables as binary vectors, where each category is assigned a unique binary vector, with only one valid position set to 1, while the others are set to 0. In the case of mRNA sequences, the four nucleotides are represented using 0 and 1. Specifically, each nucleotide is encoded as a distinct one-hot vector: A (1, 0, 0, 0), C (0, 1, 0, 0), G (0, 0, 1, 0), and T (0, 0, 0, 1), respectively. This encoding method allows nucleotide sequences to be transformed into a structure that can be efficiently handled by computational techniques.

3) Word2Vec embedding

Word2Vec is a widely used word embedding technique in natural language processing (NLP), which projects words into a continuous vector space, thereby capturing both semantic and syntactic dependencies between words [20]. Unlike conventional bag-of-words approaches, Word2Vec encodes each word as a compact, low-dimensional vector, ensuring that semantically related words are positioned close to one another in the vector space. The training of Word2Vec is typically carried out using two main models: the continuous bag of words (CBOW) model and the Skip-gram model.

In this paper, we integrate the sliding window method and the Word2Vec Skip-gram approach for feature extraction and semantic modeling of mRNA sequences. The CBOW model predicts a target word based on its surrounding context, but it tends to average contextual information, which may obscure subtle and biologically relevant signals. Such a mechanism is less compatible with the sliding window approach, as it may fail to preserve the fine-grained local dependencies essential for

accurate sequence modeling. In contrast, the Skip-gram model predicts the context given a target word, enabling it to better capture rare and informative sequence features. This characteristic makes Skip-gram particularly appropriate for integration with the sliding window method to enhance the semantic modeling of mRNA sequences. First, we use the sliding window method to extract local subsequences from the mRNA sequence. Let the given mRNA sequence be represented as:

$$S = (s_1, s_2, s_3, \dots, s_L) \quad (1)$$

where s_i is denotes the i -th nucleotide, and the sequence length is L . The sliding window method aims to extract multiple contiguous subsequences of size w from the mRNA sequence. For each window, the subsequence is represented as:

$$W_i = \{s_i, s_{i+1}, \dots, s_{i+w-1}\}, \quad \text{for } i = 1, 2, \dots, L - w + 1 \quad (2)$$

where, W_i is the subsequence from position i to position $i + w - 1$. This process generates multiple windowed subsequences to obtain the local structural patterns of the gene sequence.

After the local subsequences are extracted using the sliding window method, we apply the Skip-gram of Word2Vec to learn the word embeddings for each subsequence in the window. Specifically, given a center word w_c in the window, the model's objective is to predict its surrounding context words. This is expressed mathematically as:

$$P(w_j|w_c) = \frac{\exp(v_{w_j}^T v_{w_c})}{\sum_{w'} \exp(v_{w'}^T v_{w_c})} \quad (3)$$

where w' is the context word, and v_{w_j} and v_{w_c} represent the word vectors for the context and center words, respectively, and T denotes the dot product operation.

The objective of the model is to maximize the conditional probability of the context words given the center word, expressed as:

$$P(w_1, w_2, \dots, w_k|w_c) = \prod_{j=1}^k P(w_j|w_c) \quad (4)$$

where k is the context window size.

By combining the sliding window method and the Skip-gram model of Word2Vec, we can extract local subsequences from the gene sequence at multiple scales and learn the word embeddings for each subsequence. These word embeddings capture the semantic features of the local regions in the gene sequence, reflecting the underlying relationships between the sequences.

F. Model training and implementation

During the training of mRSubLoc, we utilized the Adam optimizer with a learning rate set to 0.0001 and a batch size of 64 to update the model's weight parameters. Additionally, gradient clipping was applied with a clipping norm threshold of 0.5 to ensure training stability. The maximum number of training epochs was set to 100.

To address the class imbalance issue in the multi-label classification task, we incorporated label-specific positive weights into the binary cross-entropy loss function. Specifically, the `pos_weight` parameter in the BCEWithLogitsLoss function was computed based on the ratio of negative to positive samples for each label in the training set.

The weight for the i -th label was calculated as follows:

$$w_i = \frac{N_i^-}{N_i^+ + \epsilon} \quad (5)$$

where N_i^+ and N_i^- represent the number of positive and negative samples for label i , respectively, and ϵ is a small constant to avoid division by zero. This weighting strategy was incorporated into the loss function:

$$\mathcal{L} = - \sum_{i=1}^9 w_i [y_i \log(\sigma(x_i)) + (1 - y_i) \log(1 - \sigma(x_i))] \quad (6)$$

where x_i is the predicted logit for class i , $y_i \in \{0, 1\}$ denotes the ground-truth label for the i -th class, and $\sigma(\cdot)$ is the sigmoid activation function. This formulation helps the model better focus on underrepresented classes during training.

Our model was developed using Python 3.9.13 and PyTorch 2.0.0+cu118. In the Windows 11 environment, we used an Intel Core i7-13700H processor, 16.0GB of memory, and NVIDIA GeForce RTX 4060 for training and evaluation.

G. Performance evaluation

To provide a comprehensive evaluation of the mRSubLoc performance, we applied five evaluation metrics that are widely adopted in multi-label classification tasks, all of which were computed based on Chou's formulas [30]. These metrics include Aiming, Coverage, Accuracy, Absolute-True, and Absolute-False. The Aiming reflects the average ratio of correctly predicted labels over the predicted labels, measuring the percentage of predicted labels that correctly match the real labels. The Coverage represents the average ratio of correctly predicted labels over the real labels, assessing the proportion of true labels successfully identified by the model. The Accuracy measures the average ratio of correctly predicted labels over the total number of labels, including correctly and incorrectly predicted labels, as well as real labels that were missed in the prediction. Compared with these three metrics, which evaluate the correctness of predictions from different perspectives, the Absolute-True is the most intuitive and stringent metric for multi-label classification. A prediction is scored as 1 only when all its labels are exactly predicted without any underprediction or overprediction; otherwise, it is scored as 0. Finally, the Absolute-False represents the proportion of incorrectly predicted labels, where a lower value indicates better prediction quality. Unlike the other four metrics, which should be maximized for better performance, the Absolute-False rate should be minimized.

III. RESULTS AND DISCUSSION

A. Performance comparison with existing methods

To assess the effectiveness of mRSubLoc, we conducted a comparative analysis against five existing multi-label prediction methods: DM3Loc, Clarion, MRSLpred, MSlocPRED and MBSCLoc. The selection of these methods is based on the following criteria: (i) availability to users through GitHub repository or an online web server; (ii) requirement of only the mRNA sequence as input; and (iii) provision of subcellular localization probabilities as output. Table 2 presents the independent test results on the testing set for existing mRNA multi-label subcellular localization models. Among the

methods, mRSubLoc demonstrated the highest performance in most of the metrics, with an Aiming of 0.7858, Coverage of 0.6212, and Accuracy of 0.6161, indicating strong prediction quality and generalization. It also achieves the best trade-off between Absolute-True (0.3070) and Absolute-False (0.1319), reflecting high reliability. Overall, mRSubLoc stands out as the most effective method for subcellular localization prediction, offering the best combination of high accuracy, reliable prediction quality, and minimal false positives.

TABLE II

COMPARE WITH EXISTING MODELS ON A TESTING SET

Methods	Aiming	Coverage	Accuracy	Absolute-True	Absolute-False
DM3Loc	0.7605	0.5327	0.4855	0.2650	0.2338
Clarion	0.7845	0.4175	0.4168	0.2823	0.2683
MRS�pred	0.6813	0.6115	0.4355	0.2532	0.2452
MSlocPRED	0.6060	0.6780	0.4503	0.2093	0.3338
MBSCLoc	0.7460	0.6584	0.5124	0.2226	0.2276
mRSubLoc	0.7858	0.6212	0.6161	0.3070	0.1319

Notes: The bold numbers indicate the best performance.

B. Effect of different features on subcellular localization

In bioinformatics research, feature encoding techniques play a critical role in enhancing the performance of RNA subcellular location. To assess the representational capacity of various features for mRNA sequences, 5-fold cross-validation is performed on the training set in this study.

TABLE III

COMPARE THE PERFORMANCE OF DIFFERENT FEATURES

Methods	Aiming	Coverage	Accuracy	Absolute-True	Absolute-False
One-hot	0.6987	0.5835	0.4851	0.1877	0.2475
Word2Vec	0.7042	0.5931	0.5061	0.1956	0.2144
RNAErnie	0.7235	0.6027	0.5345	0.2342	0.1813
One-hot+ Word2Vec	0.7278	0.6065	0.5572	0.2400	0.1743
One-hot+ RNAErnie	0.7379	0.6198	0.5704	0.2513	0.1699
Word2Vec+ RNAErnie	0.7485	0.6176	0.5818	0.2741	0.1598
One-hot+ Word2Vec+ RNAErnie	0.7758	0.6201	0.6014	0.2925	0.1463

Notes: The bold numbers indicate the best performance.

Table 3 compares the performance of different encoding methods in mRNA sublocalization task, highlighting the distinctions between different features. Among the single encoding methods, RNAErnie performs the best, achieving Aiming (0.7235), Coverage (0.6027), and Accuracy (0.5345), significantly outperforming one-hot and Word2Vec in terms of prediction consistency, coverage, and overall accuracy. This demonstrates that RNA large language models can more profoundly learn latent patterns in sequences. Combining multiple encoding methods further improves model

performance, with the "One-hot + Word2Vec + RNAErnie" combination achieving the highest Aiming (0.7758), Coverage (0.6201), and Accuracy (0.6014). Additionally, Absolute-True (0.2925) shows a significant increase, and Absolute-False (0.1463) is notably reduced. This indicates that integrating multiple encoding methods effectively leverages their respective strengths, thereby enhancing the model's predictive power and reliability.

C. Effect of subcellular localization-specific feature extraction

To validate the effectiveness of subcellular localization-specific feature extraction, 5-fold cross-validation is performed on the training set, as shown in Table 4. The model A achieves moderate results with Aiming (0.7232), Coverage (0.5963), and Accuracy (0.5434), indicating its strength in capturing local patterns but limited in handling more complex, long-range dependencies. The model B shows a slight improvement, with Aiming (0.7422), Coverage (0.6053), and Accuracy (0.5711), suggesting that its ability to model bidirectional sequential dependencies contributes to better capturing temporal relationships in the data. When TextCNN and BiLSTM are combined, the performance improves further, achieving Aiming (0.7525), Coverage (0.6132), and Accuracy (0.5846), indicating the benefit of leveraging both local and long-range dependencies. However, the integration of the multi-head self-attention mechanism with the combined TextCNN + BiLSTM leads to the most substantial performance boost, with Aiming (0.7758), Coverage (0.6201), and Accuracy (0.6014), alongside an increase in Absolute-True (0.2925) and a decrease in Absolute-False (0.1463). This demonstrates the critical role of the multi-head self-attention mechanism in enhancing the model's capability to focus on relevant features, selecting optimal features for each subcellular localization, and thereby improving prediction accuracy while reducing errors.

TABLE IV

COMPARISON OF PERFORMANCE BETWEEN DIFFERENT MODELS

Methods	Aiming	Coverage	Accuracy	Absolute-True	Absolute-False
Model A	0.7232	0.5963	0.5434	0.2502	0.1874
Model B	0.7422	0.6053	0.5711	0.2618	0.1687
Model C	0.7525	0.6132	0.5846	0.2723	0.1601
Model D	0.7758	0.6201	0.6014	0.2925	0.1463

Notes: The bold numbers indicate the best performance.

1. Model A refers to the network architecture containing only the TextCNN module.
2. Model B refers to the network architecture containing only the BiLSTM module.
3. Model C refers to the combination of the TextCNN and BiLSTM modules.
4. Model D refers to the network architecture incorporating a multi-head self-attention mechanism for extracting localization-specific features.

D. Effect of learning label relationships

To validate the effectiveness of learning the relationships between subcellular locations, 5-fold cross-validation is performed on the training set.

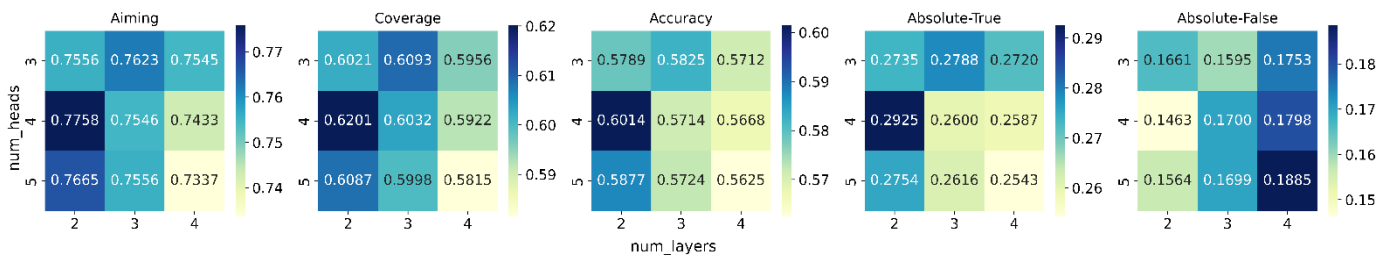


Fig. 4. Heatmap illustrating the effect of different numbers of multi-head self-attention heads and BiLSTM layers on model performance.

Fig. 3 presents a comparison between two models: one that does not account for label relationships ("No-label relationships") and another that incorporates label relationships ("mRSubLoc"). The "mRSubLoc" model outperforms the "No-label relationships" model across all metrics. Specifically, mRSubLoc and No-label relationships achieves the Aiming of 0.7758/0.7539, Coverage of 0.6201/0.6067, and Accuracy of 0.6014/0.5832, suggesting that incorporating label dependencies improves the model's capacity to target relevant outcomes and generalize effectively across various classes. Additionally, the mRSubLoc achieves significant improvements over No-label relationship methods, elevating the Absolute-True from 0.2803 to 0.2925 while reducing the Absolute-False from 0.1587 to 0.1463, indicating a higher rate of correct predictions and fewer false positives. These findings highlight the significant advantage of integrating label relationships, which contributes to more precise and dependable model predictions.

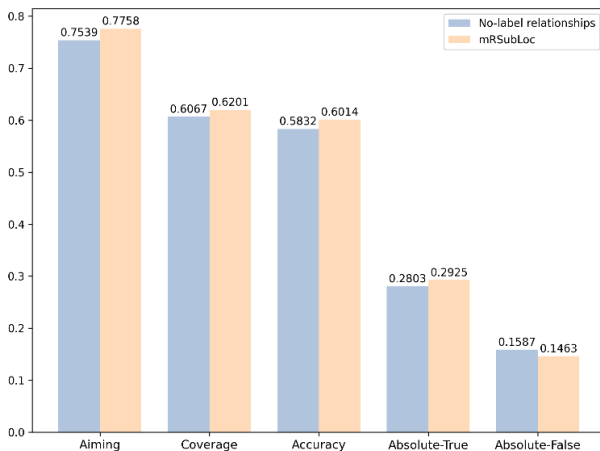


Fig. 3. The effect of learning label dependencies.

E. Effect of different numbers of multi-head attention heads and BiLSTM layers

To further validate the architecture of the model, we conducted 5-fold cross-validation on the training set to examine the effect of the number of self-attention heads and the number of BiLSTM layers on model performance. We evaluated the model by combining different numbers of self-attention heads (3, 4, 5) with different numbers of BiLSTM layers (2, 3, 4). These specific configurations were selected to achieve an optimal balance between representational capacity and computational cost, ensuring that the model remains expressive enough to capture complex contextual dependencies without

incurring excessive overhead.

As shown in Fig. 4, the horizontal axis represents the number of layers of BiLSTM, and the vertical axis represents the number of multi-head self-attention heads. It can be observed that the model achieves the best performance when the number of layers is set to 2 and the number of attention heads to 4, with Aiming of 0.7758, Coverage of 0.6201, Accuracy of 0.6014, Absolute-True of 0.2925, and Absolute-False of 0.1463. As the number of BiLSTM layers increases, the model performance declines. This may be since deeper BiLSTM layers hinder information flow and lead to the vanishing gradient problem, making it difficult for the model to effectively capture long-distance dependencies. In addition, an inappropriate number of attention heads can impair model performance by limiting semantic expressiveness or causing redundant and less focused attention on specific features.

F. Interpretation of multi-head self-attention mechanism in feature selection for subcellular localization

mRSubLoc utilizes a multi-head self-attention mechanism to obtain attention weights at the nucleic acid level, thereby identifying key motifs within mRNA sequences. These motifs, characterized by the highest information content and location-specific representations, serve as the optimal features that contribute most significantly to distinguishing different subcellular localizations. To better understand the optimal feature selection process of mRSubLoc, we selected a representative sample (NCBI ID: 396526) as a case study. Its true label is the cytoplasm, and mRSubLoc correctly predicts its subcellular localization as the cytoplasm. Kislauskis et al. [31] discovered that the repeated motif "ACACCC" in β -actin mRNA is closely associated with its localization in the cytoplasm. As shown in Fig. 5, mRSubLoc highlights the motif "ACACCC" within the sequence across different feature representations.

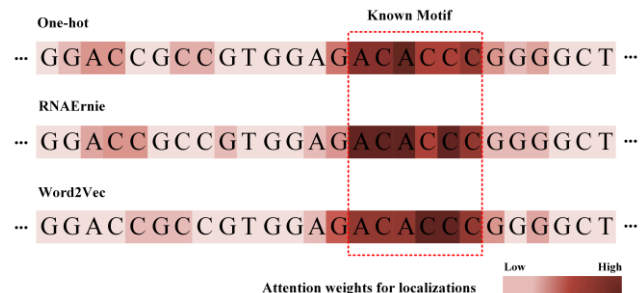


Fig. 5. Attention weight visualization.

G. Performance at different subcellular localizations

To validate the robustness of our model in predicting various

subcellular locations, we use receiver operating characteristic (ROC) curves to assess its performance. Each subcellular localization is evaluated independently by considering the presence of mRNA at that location as a binary classification task. For a given subcellular compartment, the positive samples are those mRNAs known to localize to that compartment, whereas the negative samples are those not annotated for that location. Fig. 6 presents the ROC curves for various subcellular location classifications on a testing set. The ROC curve is a graphical representation of the performance of a classification model, with the x-axis representing the false positive rate and the y-axis representing the true positive rate. The area under the curve (AUC) is a critical metric for evaluating model performance, with values closer to 1 indicating better performance. Figure. 4 shows the exosome classification stands out with a perfect AUC of 0.9453, indicating an excellent predictive ability. Other categories like nucleoplasm (AUC = 0.9353), cytoplasm (AUC = 0.9053), nucleolus (AUC = 0.9249), and membrane (AUC = 0.9167) also show strong performance. However, the cytosol category has the lowest AUC at 0.7842, suggesting room for improvement in its prediction accuracy. Overall, the model demonstrates robust performance across most subcellular locations, apart from cytosol.

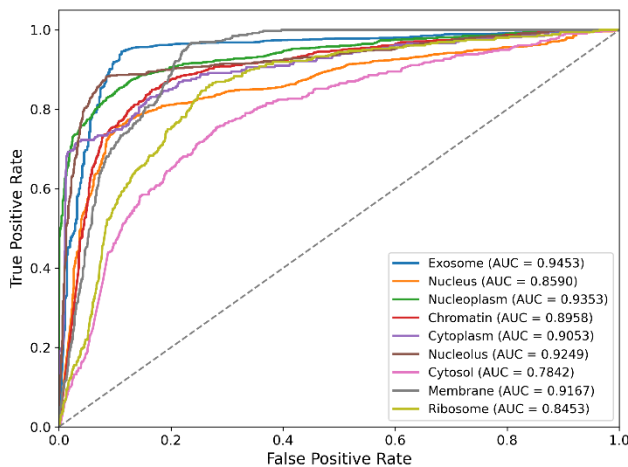


Fig. 6. The ROC curve for each subcellular localization.

H. Basic research supports clinical applications

The study of mRNA subcellular localization plays a critical role in advancing targeted clinical applications, particularly in the field of precision therapeutics. When mRNA fails to reach its designated subcellular destination, the spatial regulation of gene expression may be disrupted, protein synthesis may be impaired, and cellular signaling pathways may be compromised, thereby promoting disease development. For example, in cancer, the mislocalization of tumor suppressor or oncogenic mRNAs can lead to uncontrolled cell proliferation, while in neurodegenerative diseases such as Alzheimer's disease, the loss of proper mRNA localization in neurons can result in toxic protein aggregation and neurotoxicity [32-33].

A comprehensive understanding of the molecular mechanisms that govern mRNA localization is essential for the rational design of mRNA-based therapeutic strategies. This

knowledge enables the development of approaches that direct mRNAs to specific subcellular compartments, thereby enhancing the spatial and functional precision of treatment. For instance, the development of RNA-targeted nanocarriers enables the precise delivery of therapeutic mRNA to specific organelles, such as ribosomes, which is critical for correcting aberrant protein translation in neurodegenerative diseases, or the nucleus, which plays a pivotal role in regulating the expression of cancer-related genes, thereby significantly enhancing therapeutic efficacy [34]. In addition, abnormal patterns of mRNA localization serve as valuable indicators of disease state. In oncology, mRNA subcellular localization aids in predicting the metastatic potential or therapeutic resistance of cancer cells, thereby enabling the development of personalized treatment strategies [35].

Integrating the knowledge of mRNA subcellular localization into clinical practice provides a promising avenue for both correcting mislocalization and achieving targeted therapeutic delivery. This approach may substantially enhance treatment efficacy, minimize adverse effects, and support the development of individualized therapeutic regimens based on patients' molecular profiles. Ultimately, advances in the study of mRNA subcellular localization will lay a solid foundation for innovation in diagnostic technologies, therapeutic strategies, and precision medicine, contributing to improved clinical outcomes across a wide range of diseases.

IV. CONCLUSION

This paper proposes a novel model, mRSubLoc, for predicting multi-label messenger RNA (mRNA) subcellular localization. The contributions of this paper can be summarized as follows: (i) An RNA large language model has been introduced to holistically account for interrelationships among biomolecules, integrating deep semantic features, which significantly enhances the model's predictive accuracy and robustness. (ii) A multi-head self-attention-based method effectively extracts subcellular localization-specific features while preserving their inherent biological characteristics, enabling the model to better learn label-specific latent patterns through adaptive feature recalibration. (iii) To characterize the co-occurrence relationships among subcellular localization labels, we propose a multi-layer perceptron (MLP)-based approach that effectively captures the intricate dependencies between labels.

The results of 5-fold cross-validation on the training set indicate that RNA large language models can more effectively represent mRNA sequences. The use of a multi-head self-attention mechanism to select the optimal features at each position, and the adoption of a strategy to learn the relationships between subcellular locations, significantly enhances the model's performance. Moreover, the results on the testing set demonstrate that our approach surpasses most state-of-the-art methods across most performance metrics. This study provides an innovative solution for subcellular localization prediction in bioinformatics and opens new directions for deeper understanding of gene expression regulation and drug development.

Although the mRSubLoc exhibits strong performance, it has certain limitations. The dataset used for training the model has

an imbalance in the number of mRNA instances across different subcellular locations, which may impact the model's training process and prediction accuracy. Effectively augmenting small-sample datasets and removing inherent noise in the data to address multi-label tasks more accurately remain complex challenges. Furthermore, due to limitations in the quality and quantity of available RNA sequence data, RNA large language models still lack sufficient understanding of latent patterns in sequences, and their ability to characterize RNA sequences remains challenging.

REFERENCES

- [1] P. Fernández-Silva, J. A. Enriquez, and J. Montoya, "Replication and transcription of mammalian mitochondrial DNA," *Exp. Physiol.*, vol. 88, no. 1, pp. 41-56, 2003, doi: 10.1113/eph8802514.
- [2] N. S. Kejiou and A. F. Palazzo, "mRNA localization as a rheostat to regulate subcellular gene expression," *Wiley Interdiscip. Rev. RNA*, vol. 8, no. 4, pp. e1416, 2017, doi: 10.1002/wrna.1416.
- [3] C. Mayr, "Regulation by 3'-untranslated regions," *Annu. Rev. Genet.*, vol. 51, no. 1, pp. 171-194, 2017, doi: 10.1146/annurev-genet-120116-024704.
- [4] M. Blanco-Urrejola, A. Gaminde-Blasco, M. Gamarra, et al., "RNA Localization and Local Translation in Glia in Neurological and Neurodegenerative Diseases: Lessons from Neurons," *Cells*, vol. 10, no. 3, pp. 632, 2021, doi: 10.3390/cells10030632.
- [5] C. Meyer, A. Garzia, and T. Tuschl, "Simultaneous detection of the subcellular localization of RNAs and proteins in cultured cells by combined multicolor RNA-FISH and IF," *Methods*, vols. 118-119, pp. 101-110, 2017, doi: 10.1016/j.ymeth.2016.09.010.
- [6] F. M. Fazal, S. Han, K. R. Parker, et al., "Atlas of Subcellular RNA Localization Revealed by APEX-Seq," *Cell*, vol. 178, no. 2, pp. 473-490.e26, 2019, doi: 10.1016/j.cell.2019.05.027.
- [7] F. A. Lefebvre, N. A. L. Cody, L. P. B. Bouvrette, et al., "CeFra-seq: Systematic mapping of RNA subcellular distribution properties through cell fractionation coupled to deep-sequencing," *Methods*, vol. 126, pp. 138-148, 2017, doi: 10.1016/j.ymeth.2017.05.017.
- [8] T. Bai, J. Xie, B. Liu, and Y. Liu, "MMFmiRLocEL: A multi-model fusion and ensemble learning approach for identifying miRNA subcellular localization using RNA structure language model," *IEEE J. Biomed. Health Inform.*, pp. 1-11, 2025, doi: 10.1109/JBHI.2025.3548940.
- [9] Z. Yan, E. Lécuyer, and M. Blanchette, "Prediction of mRNA subcellular localization using deep recurrent neural networks," *Bioinformatics*, vol. 35, no. 14, pp. i333-i342, 2019, doi: 10.1093/bioinformatics/btz337.
- [10] Z. Y. Zhang, Y. H. Yang, H. Ding, et al., "Design powerful predictor for mRNA subcellular location prediction in *Homo sapiens*," *Brief. Bioinform.*, vol. 22, no. 1, pp. 526-535, 2021, doi: 10.1093/bib/bbz177.
- [11] A. Garg, N. Singhal, R. Kumar, et al., "mRNALoc: a novel machine-learning based in-silico tool to predict mRNA subcellular localization," *Nucleic Acids Res.*, vol. 48, no. W1, pp. W239-W243, 2020, doi: 10.1093/nar/gkaa385.
- [12] J. Li, L. Zhang, S. He, et al., "SubLocEP: a novel ensemble predictor of subcellular localization of eukaryotic mRNA based on machine learning," *Brief. Bioinform.*, vol. 22, no. 5, pp. bbac401, 2021, doi: 10.1093/bib/bbaa401.
- [13] G. H. Yuan, Y. Wang, G. Z. Wang, et al., "RNALight: a machine learning model to identify nucleotide features determining RNA subcellular localization," *Brief. Bioinform.*, vol. 24, no. 1, pp. bbac509, 2023, doi: 10.1093/bib/bbac509.
- [14] D. Wang, Z. Zhang, Y. Jiang, et al., "DM3Loc: multi-label mRNA subcellular localization prediction and analysis based on multi-head self-attention mechanism," *Nucleic Acids Res.*, vol. 49, no. 8, pp. e46, 2021, doi: 10.1093/nar/gkab016.
- [15] Y. Bi, F. Li, X. Guo, et al., "Clarion is a multi-label problem transformation method for identifying mRNA subcellular localizations," *Brief. Bioinform.*, vol. 23, no. 6, pp. bbac467, 2022, doi: 10.1093/bib/bbac467.
- [16] S. Choudhury, N. Bajiyi, S. Patiyal, and G. P. Raghava, "MRSLpred—a hybrid approach for predicting multi-label subcellular localization of mRNA at the genome scale," *Front. Bioinform.*, vol. 4, pp. 1341479, 2024, doi: 10.3389/fbinf.2024.1341479.
- [17] Y. Zuo, B. Zhang, W. He, et al., "MSlocPred: deep transfer learning-based identification of multi-label mRNA subcellular localization," *Brief. Bioinform.*, vol. 25, no. 6, pp. bbae504, 2024, doi: 10.1093/bib/bbae504.
- [18] B. Zhang, Y. Zuo, Z. Dai, S. Zhu, X. Liu, and Z. Deng, "MBSCLoc: Multi-label subcellular localization predict based on cluster balanced subspace partitioning method and multi-class contrastive representation learning," *IEEE J. Biomed. Health Inform.*, early access, doi: 10.1109/JBHI.2025.3537284.
- [19] N. Wang, J. Bian, Y. Li, et al., "Multi-purpose RNA language modelling with motif-aware pretraining and type-guided fine-tuning," *Nat. Mach. Intell.*, vol. 6, no. 5, pp. 548-557, 2024, doi: 10.1038/s42256-024-00836-4.
- [20] K. W. Church, "Word2Vec," *Nat. Lang. Eng.*, vol. 23, no. 1, pp. 155-162, 2017, doi: 10.1017/S1351324916000334.
- [21] B. Guo, C. Zhang, J. Liu, et al., "Improving text classification with weighted word embeddings via a multi-channel TextCNN model," *Neurocomputing*, vol. 363, pp. 366-374, 2019, doi: 10.1016/j.neucom.2019.07.052.
- [22] S. Siarni-Namini, N. Tavakoli, and A. S. Namin, "The Performance of LSTM and BiLSTM in Forecasting Time Series," in *Proc. IEEE Int. Conf. Big Data (Big Data)*, Los Angeles, CA, USA, 2019, pp. 3285-3292.
- [23] T. Cui, Y. Dou, P. Tan, et al., "RNALocate v2.0: an updated resource for RNA subcellular localization with increased coverage and annotation," *Nucleic Acids Res.*, vol. 50, no. D1, pp. D333-D339, 2022, doi: 10.1093/nar/gkab825.
- [24] W. Li and A. Godzik, "Cd-hit: a fast program for clustering and comparing large sets of protein or nucleotide sequences," *Bioinformatics*, vol. 22, no. 13, pp. 1658-1659, 2006, doi: 10.1093/bioinformatics/btl158.
- [25] A. N. Tarekegn, M. Ullah, and F. A. Cheikh, "Deep learning for multi-label learning: a comprehensive survey," *arXiv preprint arXiv:2401.16549*, 2024.
- [26] M. Akiyama and Y. Sakakibara, "Informative RNA base embedding for RNA structural alignment and clustering by deep representation learning," *NAR Genomics Bioinform.*, vol. 4, no. 1, pp. lqac012, 2022, doi: 10.1093/nargab/lqac012.
- [27] Y. Zhang, M. Lang, J. Jiang, et al., "Multiple sequence alignment-based RNA language model and its application to structural inference," *Nucleic Acids Res.*, vol. 52, no. 1, pp. e3-e3, 2024, doi: 10.1093/nar/gkad1031.
- [28] J. Chen, Z. Hu, S. Sun, et al., "Interpretable RNA foundation model from unannotated data for highly accurate RNA structure and function predictions," *arXiv preprint arXiv:2204.00300*, 2022, doi: 10.48550/arXiv.2204.00300.
- [29] J. Jia, L. Qin, and R. Lei, "DGA-5mC: A 5-methylcytosine site prediction model based on an improved DenseNet and bidirectional GRU method," *Math. Biosci. Eng.*, vol. 20, no. 6, pp. 9759-9780, 2023, doi: 10.3934/mbe.2023428.
- [30] K. C. Chou, "Some remarks on predicting multi-label attributes in molecular biosystems," *Mol. Biosyst.*, vol. 9, no. 6, pp. 1092-1100, 2013, doi: 10.1039/c3mb25555g.
- [31] E. H. Kislauskis, X. Zhu, and R. H. Singer, "Sequences responsible for intracellular localization of beta-actin messenger RNA also affect cell phenotype," *J. Cell Biol.*, vol. 127, no. 2, pp. 441-451, 1994, doi: 10.1083/jcb.127.2.441.
- [32] L. Fabbri, A. Chakraborty, C. Robert, and S. Vagner, "The plasticity of mRNA translation during cancer progression and therapy resistance," *Nat. Rev. Cancer*, vol. 21, no. 9, pp. 558-577, 2021, doi: 10.1038/s41568-021-00380-y.
- [33] J. X. Lu, Y. Wang, Y. J. Zhang, M. F. Shen, H. Y. Li, Z. Q. Yu, and G. Chen, "Axonal mRNA localization and local translation in neurodegenerative disease," *Neural Regen. Res.*, vol. 16, no. 10, pp. 1950-1957, 2021, doi: 10.4103/1673-5374.308074.
- [34] N. Naimi, H. Seyedmirzaei, Z. Hassannejad, and A. S. Khaboushan, "Advanced nanoparticle strategies for optimizing RNA therapeutic delivery in neurodegenerative disorders," *Biomed. Pharmacother.*, vol. 175, pp. 116691, 2024, doi: 10.1016/j.biopha.2024.116691.
- [35] X. Wei and M. Yang, "Cell- and subcellular organelle-targeting nanoparticle-mediated breast cancer therapy," *Front. Pharmacol.*, vol. 14, pp. 1180794, 2023, doi: 10.3389/fphar.2023.1180794.